A Combinatorial Approximation Framework for Real-Time Packet Scheduling in LoRaWAN Networks

BTP Project Report

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1 System Model and Problem Statement

1.1 Network and Hardware Model

Consider a star-of-stars LoRaWAN deployment comprising a set of distributed end-devices communicating with a central multi-channel gateway. The network operates under a cyclic hypercycle structure of duration H.

The physical transmission medium is characterized by:

- A set of N_{ch} orthogonal carrier frequencies (channels) $\mathcal{C} = \{\text{CH}_0, \text{CH}_1, \ldots, \text{CH}_{N_{\text{ch}}-1}\}$.
- A set of N_{sf} quasi-orthogonal spreading factors $\mathcal{S} = \{\text{SF}_7, \text{SF}_8, \ldots, \text{SF}_{12}\}$.

A physical communication resource is uniquely defined by a tuple $\langle \text{SF}, \text{CH} \rangle$. We model the gateway receiver capabilities as a set of $m = N_{\text{ch}} \times N_{\text{sf}}$ discrete, unrelated parallel machines:

$$\mathcal{M} = \{M_0, M_1, \ldots, M_{m-1}\} \tag{1}$$

For an 8-channel gateway operating across spreading factors SF7–SF12, the system comprises $m = 8 \times 6 = 48$ logical machines.

Hardware constraints at standard gateway digital baseband processors (e.g., Semtech SX1301/SX1302/SX1303) restrict the maximum number of concurrent demodulation paths across all channels and spreading factors to at most **8 simultaneous packets**:

$$N_{\text{demod}}^{\max} = 8 \tag{2}$$

Overlapping transmissions exceeding this threshold induce baseband receiver channel exhaustion, triggering buffer overflow, DSP pipeline saturation, and unrecoverable packet drops.

1.2 Traffic and Packet Model

Let $\mathcal{J} = \{J_0, J_1, \ldots, J_{n-1}\}$ denote the set of n real-time packets to be scheduled within the hypercycle $[0, H]$. The traffic is periodic with harmonic periods: each packet stream k possesses a period T_k such that $T_k \mid H$, generating periodic instances with release times $r_i \in [0, H)$.

Each packet $J_i \in \mathcal{J}$ is characterized by:

1. A release time $r_i \geq 0$, representing the earliest instant the end-device can initiate transmission.
2. An absolute deadline $D_i \in (r_i, H]$, clamped to the hypercycle boundary:
$$D_i \leftarrow \min(D_i, H) \tag{3}$$
3. A criticality or priority level $\Pi_i \in \mathbb{R}^+$, where higher values designate packets of greater operational importance.
4. A payload size L_i in bytes.

1.3 Time-on-Air (ToA) Matrix

The transmission duration p_{ij} of packet J_i on resource $M_j = \langle \mathrm{SF}_j, \mathrm{CH}_j \rangle$ is governed by the Semtech SX1276 physical layer specifications:

$$T_{\mathrm{sym}} = \frac{2^{\mathrm{SF}_j}}{\mathrm{BW}} \quad \text{(seconds)} \tag{4}$$

The preamble duration is $T_{\mathrm{preamble}} = (N_{\mathrm{preamble}} + 4.25) \cdot T_{\mathrm{sym}}$. The payload symbol count is given by:

$$N_{\mathrm{payload}} = 8 + \max\left(\left\lceil \frac{8L_i - 4\mathrm{SF}_j + 28 + \mathrm{CRC} - 20\mathrm{IH}}{4(\mathrm{SF}_j - 2\mathrm{DE})} \right\rceil (\mathrm{CR} + 4), 0 \right) \tag{5}$$

where CRC = 16 bits, IH = 0 (explicit header enabled), CR = 1 (coding rate 4/5), and DE = 1 if $T_{\mathrm{sym}} > 16$ ms (Low Data Rate Optimization), and 0 otherwise.

The entries of the $n \times m$ Time-on-Air (ToA) matrix $P = [p_{ij}]$ are discretized into integer milliseconds:

$$p_{ij} = \begin{cases} \lceil (T_{\mathrm{preamble}} + N_{\mathrm{payload}} \cdot T_{\mathrm{sym}}) \times 1000 \rceil, & \text{if packet } J_i \text{ is eligible on } M_j \\ \infty, & \text{otherwise} \end{cases} \tag{6}$$

We define the maximum processing time and the number of admissible edges as:

$$U = \max_{i \in \mathcal{J}, j \in \mathcal{M}} \{p_{ij} \mid p_{ij} < \infty\}, \quad A = |\{(i,j) \mid p_{ij} < \infty\}| \leq n \cdot m \tag{7}$$

1.4 LoRaWAN Class A Energy Consumption Model

Energy efficiency is paramount for battery-powered LoRaWAN IoT end-devices. Following the established analytical frameworks of Casals et al. [1] and Bouguera et al. [2], the energy consumption per transmission cycle E_{cycle} encompasses all operational radio and microcontroller states:

$$E_{\text{cycle}} = V_{\text{dd}} \cdot \sum_{s \in \mathcal{S}_{\text{states}}} (I_s \cdot t_s) \tag{8}$$

where $\mathcal{S}_{\text{states}} = \{\text{MCU}, \text{Sensor}, \text{TX}, \text{Wait1}, \text{RX1}, \text{Wait2}, \text{RX2}, \text{Sleep}\}$. Specifically:

$$E_{\text{tx}}(i,j) = V_{\text{dd}} \cdot I_{\text{tx}} \cdot p_{ij} \tag{9}$$

Because the symbol duration $T_{\text{sym}} = 2^{\text{SF}}/\text{BW}$ grows exponentially with Spreading Factor, Time-on-Air and transmit energy scale sharply. For instance, for a 10-byte application payload (23 bytes total PHY payload with LoRaWAN MAC overhead), transmitting at SF_{12} consumes 233.7 mJ compared to only 27.4 mJ at SF_7 ($\approx 8.5\times$ energy inflation).

Consequently, our scheduling objective is two-fold:

1. **Makespan Guarantee:** Bound channel congestion such that $C_{\max}^{\text{mach}}(\alpha) \leq 2 \cdot \text{OPT}(P)$.
2. **Energy Minimization:** Minimize total fleet radio energy $\sum_i E_{\text{cycle}}(i, \alpha(i))$ by aggressively downshifting packets to lower Spreading Factors without violating the makespan upper bound.

1.5 Problem Formulation

An assignment of packets to resources is defined by a mapping function:

$$\alpha : \mathcal{J} \to \mathcal{M} \tag{10}$$

The total workload on resource M_j under assignment α is the cumulative transmission duration of all packets assigned to j:

$$\delta_j(P, \alpha) = \sum_{i \in \mathcal{J}, \alpha(i)=j} p_{ij} \tag{11}$$

To avoid ambiguity between scheduling metrics, we explicitly distinguish:

Definition 1 (Machine Load Makespan)**.** *The maximum channel workload across all parallel resources:*

$$C_{\max}^{mach}(\alpha) = \max_{j \in \mathcal{M}} \delta_j(P, \alpha), \quad OPT(P) = \min_{\alpha} C_{\max}^{mach}(\alpha) \tag{12}$$

Definition 2 (Timeline Calendar Span)**.** *The latest chronological packet completion time along the hypercycle timeline:*

$$C_{\max}^{time} = \max_{i \in \mathcal{J}_{sched}} f_i \leq H \tag{13}$$

The combinatorial 2-approximation theorem applies specifically to minimizing the machine load makespan $C_{\max}^{\text{mach}}(\alpha)$. In addition to load balancing, each resource M_j represents a single non-preemptive communication channel. To satisfy real-time constraints, the set of packets $\mathcal{J}_j = \{J_i \in \mathcal{J} \mid \alpha(i) = j\}$ assigned to resource M_j must be chronologically schedulable without deadline violations and without violating the gateway's concurrent demodulation threshold ($N_{\text{demod}}^{\max} = 8$).

2 Combinatorial 2-Approximation on Unrelated Machines

We apply the purely combinatorial polynomial-time 2-approximation algorithm of Gairing, Monien, and Woclaw [3] for scheduling on unrelated parallel machines ($R||C_{\max}$).

2.1 Generalized Unsplittable Flow Formulation

Given a candidate makespan $w \in \mathbb{Z}^+$, we construct a bipartite directed network with node set $V = \mathcal{J} \cup \mathcal{M}$, where job nodes $\mathcal{J}$ represent packets and machine nodes $\mathcal{M}$ represent resources. Each job node $i \in \mathcal{J}$ possesses a supply of 1 unit. An edge from job i to machine j exists if and only if $p_{ij} \leq w$, with unit capacity and multiplicative gain $\mu_{ij} = p_{ij}$.

A generalized unsplittable flow routes the entire 1 unit of supply from each job node i to exactly one machine j, creating an accumulated load $\delta_j(P, \alpha)$ on machine j. An assignment α is termed *w-feasible* if $p_{i,\alpha(i)} \leq w$ for all $i \in \mathcal{J}$.

2.2 Residual Network and Machine Partitioning

For a current w-feasible assignment α, we define the directed residual graph $G_\alpha(w) = (V, E_\alpha(w))$ with edge set $E_\alpha(w) = E_\alpha^1 \cup E_\alpha^2$:

- **Backward Edges (E_α^1):** For each assigned job $\alpha(i) = j$, add directed edge (j, i) of capacity 1 and gain $\mu_{ji} = 1/p_{ij}$. This represents removing job i from machine j.
- **Forward Edges (E_α^2):** For each unassigned pair where $\alpha(i) \neq j$ and $p_{ij} \leq w$, add directed edge (i, j) of capacity 1 and gain $\mu_{ij} = p_{ij}$. This represents reassigning job i to machine j.

Given the current loads $\delta_j(P, \alpha)$, the machines are partitioned into three mutually disjoint sets:

Definition 3 (Machine Partitioning)**.**

$$\mathcal{M}^-(\alpha) = \{j \in \mathcal{M} \mid \delta_j(P, \alpha) \leq w\} \quad \textit{(Underloaded / Sinks)} \tag{14}$$

$$\mathcal{M}^0(\alpha) = \{j \in \mathcal{M} \mid w + 1 \leq \delta_j(P, \alpha) \leq 2w\} \quad \textit{(Medium / Transit)} \tag{15}$$

$$\mathcal{M}^+(\alpha) = \{j \in \mathcal{M} \mid \delta_j(P, \alpha) \geq 2w + 1\} \quad \textit{(Overloaded / Sources)} \tag{16}$$

If $\mathcal{M}^+(\alpha) = \emptyset$, every machine satisfies $\delta_j(P, \alpha) \leq 2w$, which yields a valid 2-approximation provided $\mathrm{OPT}(P) \geq w$.

2.3 Gain Rounding and Truemper's Logarithmic Transformation

To apply minimum-cost flow techniques without introducing numerical instability, edge gains are geometrically rounded down to integer powers of a base $b > 1$:

$$b = 1 + \frac{1}{m} \tag{17}$$

For each edge (i, j), the cost is defined by the discrete logarithm:

$$c_{ij} = \lfloor \log_b(p_{ij}) \rfloor \in \mathbb{Z} \tag{18}$$

The rounded gain is $\gamma_{ij} = b^{c_{ij}}$. For reverse edges (j, i), antisymmetry is enforced: $c_{ji} = -c_{ij}$ and $\gamma_{ji} = 1/\gamma_{ij} = b^{-c_{ij}}$.

Crucially, the rounding ensures:

$$\gamma_{ij} \leq p_{ij} \leq b \cdot \gamma_{ij} \tag{19}$$

The relative error introduced per edge is strictly bounded by $b - 1 = 1/m$.

2.4 Node Potentials and Reduced Cost Optimality

A node potential function $\pi : V \to \mathbb{R}$ is maintained across all nodes. For any directed edge (u, v), the reduced cost is defined as:

$$c^\pi_{uv} = c_{uv} - \pi(u) + \pi(v) \tag{20}$$

The Primal-Dual algorithm preserves the reduced cost optimality condition:

$$c^\pi_{uv} \geq 0 \quad \forall (u, v) \in E_\alpha(w) \tag{21}$$

This condition guarantees that no negative-cost cycles exist in the residual network with respect to edge costs c.

3 Algorithmic Mechanics and Theoretical Proofs

The approximation framework operates via two nested algorithmic procedures: the inner loop, *Unsplittable-Blocking-Flow*, and the outer loop, *Unsplittable-Truemper*.

3.1 Inner Loop: Unsplittable-Blocking-Flow

The inner procedure pushes discrete jobs along alternating paths from overloaded machines $\mathcal{M}^+$ toward underloaded machines $\mathcal{M}^-$ on a shortest-path level graph.

Let $G'_\alpha(w)$ be an admissible subgraph consisting of edges with zero reduced cost ($c^\pi_{uv} = 0$). A height function $h : V \to \mathbb{N} \cup \{\infty\}$ is computed, where $h(v)$ represents the shortest hop distance from node v to any sink in $\mathcal{M}^-$ in $G'_\alpha(w)$. An edge (u, v) is *admissible* if:

$$h(u) = h(v) + 1 \tag{22}$$

Definition 4 (Helpful Machine and Helpful Path)**.** *A machine $j \in \mathcal{M}^+$ is a **helpful machine** if it can reassign its first admissible job $s(j)$ while keeping its remaining load at or above $w + 1$:*

$$\delta_j(P, \alpha) \geq w + 1 + p_{s(j),j} \tag{23}$$

*A **helpful path** is an alternating sequence of nodes $v_0, v_1, \ldots, v_r$ where even indices $v_{2i} \in \mathcal{M}$ are machines and odd indices $v_{2i+1} \in \mathcal{J}$ are jobs, such that:*

1. *$v_0 \in \mathcal{M}^+$ is a helpful machine of minimum height.*
2. *Every edge (v_i, v_{i+1}) is admissible: $h(v_i) = h(v_{i+1}) + 1$.*
3. *For all intermediate machines ($0 < i < r/2$):*

$$w + 1 \leq \delta_{v_{2i}} + p_{s(v_{2i-2}),v_{2i}} - p_{s(v_{2i}),v_{2i}} \leq 2w \tag{24}$$

4. *For the terminal machine $v_r \in \mathcal{M}^-$:*

$$\delta_{v_r} + p_{s(v_{r-2}),v_r} \leq 2w \tag{25}$$

Reassigning jobs along a helpful path shifts one job from each machine to the next. The load of v_0 decreases but remains $\geq w + 1$; intermediate machines remain strictly within $\mathcal{M}^0$; and the load of v_r increases but never exceeds $2w$.

Lemma 1 (Load Invariance [3])**.** *Let β be the assignment returned by Algorithm 1 (Unsplittable-Blocking-Flow). Then:*

1. *If $j \in \mathcal{M}^-(\alpha)$, then $\delta_j(P,$ $beta$) $\geq \delta_j(P, \alpha)$ (sink loads never decrease).*
2. *If $j \in \mathcal{M}^0($ $alpha$), then $w + 1 \leq \delta_j(P, \beta) \leq 2w$ (medium machines stay medium).*
3. *If $j \in \mathcal{M}^+($ $alpha$), then $\delta_j(P, \beta) \leq \delta_j(P, \alpha)$ (source loads never increase).*

Theorem 1 (Complexity of Inner Loop [3])**.** *The inner loop terminates in at most m phases. Since each phase requires a Breadth-First Search (BFS) of complexity $O(A)$, the total running time of Unsplittable-Blocking-Flow is $O(mA)$. Upon termination, no path exists from $\mathcal{M}^+(\beta)$ to $\mathcal{M}^-(\beta)$ in $G'_\beta(w)$.*

Algorithm 1 Unsplittable-Blocking-Flow($\alpha, G'_\alpha(w), P, w$)

Input: Assignment α, admissible graph $G'_\alpha(w)$, processing matrix P, parameter w
Output: Updated assignment β

1. Compute height labels $h(v)$ for all $v \in V$ as shortest hop distance to $\mathcal{M}^-$ in $G'_\alpha(w)$ via backward BFS.
2. **while** $\mathcal{M}^- \neq \emptyset$ **and** $\exists j \in \mathcal{M}^+$ with $h(j) < \infty$ **do**
3. Identify helpful machines: $\mathcal{H} \leftarrow \{j \in \mathcal{M}^+ \mid h(j) < \infty, \delta_j \geq w + 1 + p_{s(j),j}\}$.
4. **if** $\mathcal{H} = \emptyset$ **then break**
5. Select a helpful machine $v_0 \in \mathcal{H}$ of minimum height.
6. Construct helpful path induced by v_0.
7. Push jobs along the helpful path: update α, machine loads δ, and admissible graph edges.
8. Recompute height labels h via BFS.
9. **end while**
10. **return** α

3.2 Outer Loop: Unsplittable-Truemper

The outer loop maintains node potentials $\pi(v)$ and iteratively constructs the admissible shortest-path graph for the inner loop using reduced-cost Dijkstra on the reversed residual graph.

Theorem 2 (Lower Bound Certificate [3])**.** *If Algorithm 2 terminates with $\mathcal{M}^+ \neq \emptyset$ due to the potential ceiling $\max_{j \in \mathcal{M}^+} \pi(j) \geq \log_b(m)$ or because no path connects $\mathcal{M}^+$ to $\mathcal{M}^-$, then no fractional or unsplittable assignment of makespan $\leq w$ exists; hence $OPT(P) \geq w + 1$.*

3.3 Binary Search and 2-Approximation Theorem

Theorem 3. *Algorithm 2 combined with binary search computes an assignment α satisfying:*

$$C_{\max}^{mach}(\alpha) \leq 2 \cdot OPT(P) \tag{26}$$

in total time $O(m^2 A \log m \log(w_{\max} - w_{\min}))$.

Proof. We perform binary search over $w \in [w_{\min}, w_{\max}]$, where $w_{\min} = \max_i \min_j p_{ij}$ and $w_{\max} = 2\lceil \sum_i \min_j p_{ij}/m \rceil + 2 \max_{i,j} p_{ij}$. We locate the critical threshold w^* such that Algorithm 2 with parameter $w^* - 1$ terminates with $\mathcal{M}^+ \neq \emptyset$ (proving $\mathrm{OPT}(P) \geq w^*$ by Theorem 2), while with parameter w^* it terminates with $\mathcal{M}^+ = \emptyset$.

When $\mathcal{M}^+ = \emptyset$, every machine satisfies $\delta_j(P, \alpha) \leq 2w^*$. Thus:

$$C_{\max}^{\mathrm{mach}}(\alpha) \leq 2w^* \leq 2 \cdot \mathrm{OPT}(P) \tag{27}$$

The binary search requires at most $O(\log(w_{\max} - w_{\min}))$ calls. With each call costing $O(m^2 A \log m)$, the overall running time is $O(m^2 A \log m \log(w_{\max} - w_{\min}))$. □

Algorithm 2 Unsplittable-Truemper(α, P, C, w)

Input: Initial w-feasible assignment α, matrices P and C, parameter w
Output: Final assignment β, or $\emptyset$ if $\mathrm{OPT}(P) > w$

1. Initialize potentials: $\pi(v) \leftarrow 0$ for all $v \in \mathcal{M}$; $\pi(i) \leftarrow C_{i,\alpha(i)}$ for all $i \in \mathcal{J}$.
2. **while** $\mathcal{M}^+ \neq \emptyset$ **do**
3. **if** $\max_{j \in \mathcal{M}^+} \pi(j) \geq \log_b(m)$ **then return** $\emptyset$ *// Potential ceiling reached*
4. Compute shortest distances $d(v)$ from all nodes to sinks $\mathcal{M}^-$ in reversed residual graph via Dijkstra using reduced costs c^{π}_{uv}.
5. **if** $\forall j \in \mathcal{M}^+ : d(j) = \infty$ **then return** $\emptyset$ *// No path from $\mathcal{M}^+$ to $\mathcal{M}^-$*
6. Update potentials: $\pi(v) \leftarrow \pi(v) + d(v)$ for all $v \in V$ with $d(v) < \infty$.
7. Construct admissible graph $G'_\alpha(w)$ containing edges with $c^{\pi}_{uv} = 0$.
8. $\alpha_{\mathrm{prev}} \leftarrow \alpha$.
9. $\alpha \leftarrow$ Unsplittable-Blocking-Flow($\alpha, G'_\alpha(w), P, w$).
10. **if** $\alpha = \alpha_{\mathrm{prev}}$ and all $d(j) = 0$ on reachable sources **then return** $\emptyset$.
11. **end while**
12. **return** α

4 LoRaWAN Hypercycle Scheduling with Hardware Demodulation Constraints

4.1 Gateway Hardware Demodulation Constraint

In standard LoRaWAN deployments, the central gateway is equipped with multi-channel digital baseband processors (such as Semtech SX1301/SX1302/SX1303). Although the gateway listens across all spreading factors $\mathcal{S} = \{\mathrm{SF}_7, \ldots, \mathrm{SF}_{12}\}$ defining $m = N_{\mathrm{ch}} \times N_{\mathrm{sf}} = 48$ logical communication pipes $\mathcal{M}$, the physical modem architecture possesses a finite pool of demodulation paths, strictly capping the maximum number of simultaneous packet receptions across all channels to at most **8 concurrent demodulations**:

$$N^{\max}_{\mathrm{demod}} = 8 \tag{28}$$

When more than 8 packets overlap in time across all frequency channels and spreading factors, the baseband modem experiences receiver channel exhaustion, causing packet drops due to hardware buffer overflow and DSP pipeline saturation. Therefore, any physically realizable schedule must satisfy two complementary conditions:

1. **Single-Channel Non-Preemption:** On any individual resource $M_j = \langle \mathrm{SF}_j, \mathrm{CH}_j \rangle$, transmissions cannot overlap in time.
2. **Global Modem Demodulation Limit:** Across all m resources, the total number of simultaneous active transmissions at any continuous instant $t \in [0, H]$ must not exceed 8.

Let $[s_i, f_i)$ represent the scheduled transmission interval of packet J_i on its assigned resource $M_j = \alpha(i)$, where $f_i = s_i + p_{ij}$. The global gateway

demodulation occupancy at instant t is:

$$\Omega(t) = sum_{i \in \mathcal{J}_{\text{sched}}} \mathbb{I}(s_i \leq t < f_i) \tag{29}$$

The global hardware constraint requires:

$$\Omega(t) \leq 8 \quad \forall t \in [0, H] \tag{30}$$

4.2 Per-Channel Earliest Deadline First (EDF) Timeline

For each individual machine $M_j \in \mathcal{M}$, the assigned packets $\mathcal{J}_j = \{J_i \in \mathcal{J}_{\text{cand}} \mid \alpha(i) = j\}$ are sorted in ascending order of deadlines (Earliest Deadline First):

$$D_{(1)} \leq D_{(2)} \leq \cdots \leq D_{(k)} \tag{31}$$

To maintain $\Omega(t) \leq 8$ without expensive continuous sweeps, we implement a Segment Tree with lazy propagation over $[0, H]$. A min-heap priority queue coordinates multi-channel progression:

$$s_i = \max\left(\text{cursor}_j, r_i\right) \tag{32}$$

$$f_i = s_i + p_{ij} \tag{33}$$

The demodulation peak is checked in $O(\log H)$ time via range_max$(s_i, f_i - 1)$. If peak ≥ 8 or $f_i > D_i$, the packet is flagged as an offender. Otherwise, the occupancy is updated via range_add$(s_i, f_i - 1, +1)$ in $O(\log H)$ time, and $\text{cursor}_j \leftarrow f_i$.

4.3 Phase 1.5: Energy-Optimal Capacity-Constrained Downshifting

While Phase 1 produces a balanced assignment α satisfying $C_{\max}^{\text{mach}}(\alpha) \leq 2 \cdot \text{OPT}(P)$, it optimizes makespan without explicit regard to radio energy. Because energy is heavily penalized on higher Spreading Factors ($E_{i,\text{SF12}} \gg E_{i,\text{SF7}}$), many packets can be migrated to lower Spreading Factors without exceeding the machine capacity budget.

We formulate the post-Phase 1 energy optimization problem as follows:

$$\min_{\alpha^*} \quad \sum_{i \in \mathcal{J}_{\text{cand}}} E_{\text{cycle}}(i, \alpha^*(i)) \tag{34}$$

$$\text{subject to} \quad \sum_{i:\alpha^*(i)=j} p_{ij} \leq 2w^* \quad \forall j \in \mathcal{M} \tag{35}$$

$$p_{i,\alpha^*(i)} \leq w^* \quad \forall i \in \mathcal{J}_{\text{cand}} \tag{36}$$

Theorem 4 (Preservation of the 2-Approximation Upper Bound under Phase 1.5)**.** *Let α^* be any assignment produced by Phase 1.5 satisfying constraint* (35)*. Then:*

$$C_{\max}^{mach}(\alpha^*) \leq 2 \cdot OPT(P) \tag{37}$$

Furthermore, the total radio energy is monotonically non-increasing:

$$\sum_i E_{cycle}(i, \alpha^*(i)) \leq \sum_i E_{cycle}(i, \alpha(i)) \tag{38}$$

Proof. By Theorem 2, the critical threshold found by binary search provides a lower bound on the optimum: $\mathrm{OPT}(P) \geq w^*$. Constraint (35) enforces that no machine $j \in \mathcal{M}$ incurs total processing time exceeding $2w^*$. Therefore:

$$C_{\max}^{\mathrm{mach}}(\alpha^*) = \max_{j \in \mathcal{M}} \delta_j(P, \alpha^*) \leq 2w^* \leq 2 \cdot \mathrm{OPT}(P) \tag{39}$$

Because the initial assignment α from Phase 1 already satisfies $\delta_j(P, \alpha) \leq 2w^*$, α is a feasible solution to the constrained optimization problem. Since Phase 1.5 accepts only transitions that yield non-negative energy reduction ($\Delta E \geq 0$), the total energy cannot increase. □

4.4 Complete Main Algorithm Pseudocode

The complete end-to-end scheduling framework is formalized in Algorithm 3. It integrates makespan minimization via Unsplittable-Truemper, energy-optimal capacity-constrained downshifting (Phase 1.5), per-channel EDF timeline construction, Segment-Tree-driven 8-demodulation verification, and priority-conscious packet dropping.

4.5 Hierarchical Cyclic Executive Architecture and Inter-Cycle Carryover Buffer

In real-world LoRaWAN deployments, traffic is inherently periodic and runs indefinitely across repeating major hypercycles $H = \mathrm{lcm}(T_k)$. However, scheduling over a monolithic hypercycle horizon can incur latency accumulation for high-frequency streams and lacks adaptive recovery when transient RF bursts occur. To solve this, our framework implements a hierarchical **Cyclic Executive Architecture** supporting both uniform (homogeneous) and variable-length (heterogeneous) minor frames:

1. **Macro and Micro Time Horizon Partitioning:** The major hypercycle H is partitioned into K minor operational dispatch frames of duration τ_k ($k \in \{0, 1, \ldots, K-1\}$):
   - **Homogeneous (Uniform) Frames:** All minor cycles share an identical duration $\tau_k = \tau$, where $H = K \cdot \tau$ and $H \pmod{\tau} = 0$.
   - **Heterogeneous (Variable-Length) Frames:** Minor cycle durations may differ ($\tau_0, \tau_1, \ldots, \tau_{K-1}$ with $\tau_k > 0$) to align with multi-rate physical sensor activation epochs, subject to the partition invariant:

$$\sum_{k=0}^{K-1} \tau_k = H \tag{40}$$

Algorithm 3 LoRaWAN-Hypercycle-Scheduler($\mathcal{J}, \mathcal{M}, P, E, H, N_{\text{demod}}^{\max}$)

Input: Packet set $\mathcal{J}$ with release times r_i, deadlines D_i, priorities Π_i; Resource set $\mathcal{M}$ ($m = N_{\text{ch}} \times N_{\text{sf}}$ pairs); ToA matrix P; Energy matrix E; Hypercycle H; Gateway demodulation limit $N_{\text{demod}}^{\max} = 8$

Output: Feasible schedule $\mathcal{S} = \{(J_i, \alpha(i), s_i, f_i)\}$, Dropped packet set $\mathcal{J}_{\text{drop}}$

$\mathcal{J}_{\text{cand}} \leftarrow \mathcal{J}$, $\mathcal{J}_{\text{drop}} \leftarrow \emptyset$.
Clamp deadlines: $D_i \leftarrow \min(D_i, H)$ for all $J_i \in \mathcal{J}_{\text{cand}}$.
while $\mathcal{J}_{\text{cand}} \neq \emptyset$ **do**
  // Phase 1: Unrelated Machine Makespan Minimization
  Execute binary search over $w \in [w_{\min}, w_{\max}]$ using Unsplittable-Truemper:
    Find assignment $\alpha : \mathcal{J}_{\text{cand}} \to \mathcal{M}$ and critical threshold w^*.
    Compute machine makespan $M_{\text{span}} \leftarrow \max_{j \in \mathcal{M}} \sum_{i:\alpha(i)=j} p_{ij}$.
  if $M_{\text{span}} > H$ **then**
    Identify candidate packet with lowest priority: $J^* \leftarrow \arg\min_{J_i \in \mathcal{J}_{\text{cand}}} (\Pi_i, -D_i, i)$.
    $\mathcal{J}_{\text{drop}} \leftarrow \mathcal{J}_{\text{drop}} \cup \{J^*\}$, $\mathcal{J}_{\text{cand}} \leftarrow \mathcal{J}_{\text{cand}} \setminus \{J^*\}$.
    continue *// Restart Phase 1 with reduced candidate set*
  end if
  // Phase 1.5: Energy-Optimal Capacity-Constrained Downshifting
  $\alpha \leftarrow$ Energy-Optimal-Downshift($\mathcal{J}_{\text{cand}}, \mathcal{M}, P, E, \alpha, w^*, H$).
  // Phase 2: Chronological EDF Placement with Segment Tree Demodulation Tracking
  while $\mathcal{J}_{\text{cand}} \neq \emptyset$ **do**
    Initialize lazy Segment Tree over $[0, H]$ and channel cursors $\text{cursor}_j \leftarrow 0$.
    Simulate chronological placement across all channels using min-heap priority queue.
    Identify deadline violators $\mathcal{V}_{\text{dead}}$ and demodulation peak violators $\mathcal{V}_{\text{demod}}$.
    if $\mathcal{V}_{\text{dead}} = \emptyset$ **and** $\mathcal{V}_{\text{demod}} = \emptyset$ **then**
      Construct schedule $\mathcal{S} \leftarrow \{(J_i, \alpha(i), s_i, f_i) \mid J_i \in \mathcal{J}_{\text{cand}}\}$.
      return $(\mathcal{S}, \mathcal{J}_{\text{drop}})$
    end if
    $\mathcal{V}_{\text{offending}} \leftarrow \mathcal{V}_{\text{dead}}$ **if** $\mathcal{V}_{\text{dead}} \neq \emptyset$ **else** $\mathcal{V}_{\text{demod}}$.
    Drop lowest priority offender: $J^* \leftarrow \arg\min_{J_i \in \mathcal{V}_{\text{offending}}} (\Pi_i, -D_i, i)$.
    $\mathcal{J}_{\text{drop}} \leftarrow \mathcal{J}_{\text{drop}} \cup \{J^*\}$, $\mathcal{J}_{\text{cand}} \leftarrow \mathcal{J}_{\text{cand}} \setminus \{J^*\}$.
    Prune J^* from α and retry Phase 2 directly without re-running Phase 1.
  end while
end while
return $(\emptyset, \mathcal{J})$ *// Infeasible instance*

In both cases, frame boundaries are defined by $t_0 = 0$ and $t_{k+1} = t_k + \tau_k$, establishing mutually disjoint execution windows $[t_k, t_{k+1})$ for all $k \in \{0, \ldots, K-1\}$.

2. **Inter-Cycle Carryover Buffer with Slack Viability Filtering:** Under periodic LoRaWAN traffic, deadlines are governed by the stream pe-

riod: $D_i = r_i + T_i$. When a sensor stream has period $T_i > \tau_k$ (e.g., $T_i \in \{20, 50, 100\}$ s with $\tau_k \in [10, 25]$ s), a packet dropped in minor frame k due to transient demodulator or channel contention does not immediately constitute a deadline violation.

A dropped packet J_i is eligible for deferral into minor frame $k+1$ if and only if its earliest possible finish time in frame $k+1$ is strictly within its hard deadline:

$$t_{k+1} + \min_{j \in \mathcal{M}} p_{ij} \leq D_i \quad \text{and} \quad h_i + 1 \leq h_{\max} \quad \text{and} \quad t_{k+1} < H \tag{41}$$

where t_{k+1} is the start time of frame $k+1$, and h_i denotes the number of carryover hops traversed by J_i. If this condition is violated, the packet cannot meet its deadline on any machine and is discarded as an unrecoverable hard deadline miss.

3. **Starvation Prevention via Priority Escalation:** To guarantee timely service and bound scheduling jitter, carried packets receive dynamic priority boosting in subsequent cycles:

$$\Pi_i' = \Pi_i + h_i \cdot \Delta_\Pi \tag{42}$$

ensuring that deferred packets take precedence over freshly arriving low-priority traffic.

Theorem 5 (Preservation of the 2-Approximation Guarantee and RF Invariants under Homogeneous and Heterogeneous Cyclic Scheduling)**.** *For any minor frame $k \in \{0, \ldots, K-1\}$ of duration τ_k with candidate packet set $\mathcal{J}^{(k)}$ (comprising new arrivals released in $[t_k, t_{k+1})$ and viable carried-over packets), the cyclic executive scheduler guarantees:*

1. ***Per-Frame 2-Approximation:*** *The maximum machine load satisfies:*

$$C_{\max}^{(k)} \leq 2w_k^* \leq 2 \cdot OPT_k \leq \tau_k \tag{43}$$

where OPT_k is the optimum makespan for $\mathcal{J}^{(k)}$ on machines $\mathcal{M}$ within frame horizon τ_k.

2. ***Global RF Invariant Preservation:*** *Because minor frame timelines are mutually disjoint ($[t_k, t_{k+1}) \cap [t_l, t_{l+1}) = \emptyset$ for $k \neq l$), per-channel non-preemption and gateway demodulation occupancy:*

$$\Omega(t) \leq N_{demod}^{\max} = 8, \quad \forall t \in [0, H] \tag{44}$$

hold continuously and unconditionally across the entire global hypercycle.

Proof. In each minor frame k, candidate packets $\mathcal{J}^{(k)}$ are assigned via Algorithm 3 over the local horizon τ_k. By Theorem 3, Phase 1 produces an assignment with load bounded by $2w_k^* \leq 2 \cdot \mathrm{OPT}_k$. By Theorem 4, Phase 1.5

strictly enforces that no machine load exceeds $2w_k^*$. Because all transmissions assigned to frame k terminate before t_{k+1}, no RF transmission overlaps or demodulation spillover can occur across frame boundaries. Thus, global physical constraints and the 2-approximation makespan bound are preserved across all frames, whether durations τ_k are uniform or variable. □

5 A Priori Feasibility and Schedulability Analysis via Upper Bound Duality

In real-world mission-critical IoT systems, network operators require formal verification that real-time packet deadlines will be satisfied before network deployment. While empirical simulations provide valuable experimental insights, they cannot guarantee correctness across all operational conditions. To address this, we establish an **analytical, simulation-free feasibility theorem** by leveraging upper-bound duality between our combinatorial engine and the classical Linear Programming relaxation of Lenstra, Shmoys, and Tardos (LST 1990) [4].

5.1 The Lenstra–Shmoys–Tardos (1990) LP Relaxation and Rounding Bound

Consider the parametric continuous linear program $\mathrm{LP}(T)$ for a guessed makespan T and fleet radio energy budget $\mathcal{E}_{\text{budget}}$:

$$\sum_{j \in \mathcal{M}} x_{ij} = 1 \qquad \forall i \in \mathcal{J} \tag{45}$$

$$\sum_{i \in \mathcal{J}} x_{ij} \cdot p_{ij} \leq T \qquad \forall j \in \mathcal{M} \tag{46}$$

$$x_{ij} = 0 \qquad \forall (i,j) \text{ with } p_{ij} > T \text{ or } p_{ij} > D_i \tag{47}$$

$$\sum_{i \in \mathcal{J}} \sum_{j \in \mathcal{M}} x_{ij} \cdot E_{ij} \leq \mathcal{E}_{\text{budget}} \tag{48}$$

$$x_{ij} \geq 0 \qquad \forall i \in \mathcal{J}, j \in \mathcal{M} \tag{49}$$

Theorem 6 (LST Bipartite Matching Rounding Bound [4, 5])**.** *If LP(T) is feasible, any basic feasible solution (extreme point) x^* can be rounded in polynomial time via maximum bipartite matching on the fractional graph $G_F = (\mathcal{J}_F \cup \mathcal{M}_F, E_F)$ into an integral assignment $\hat{x} \in \{0,1\}^{n \times m}$ such that:*

$$\sum_{i \in \mathcal{J}} \hat{x}_{ij} p_{ij} \leq T + \max_{i : x^*_{ij} > 0} p_{ij} \leq 2T, \quad \forall j \in \mathcal{M} \tag{50}$$

and $\sum_{i,j} \hat{x}_{ij} E_{ij} \leq \mathcal{E}_{budget}$.

Proof. The constraint matrix of $\mathrm{LP}(T)$ possesses $n+m+1$ functional constraints. Consequently, any basic feasible solution has at most $n+m+1$ non-zero variables

($x^*_{ij} > 0$). Because each of the n jobs requires $\sum_j x_{ij} = 1$, at most $m + 1$ jobs receive fractional assignments. Standard cycle canceling eliminates cycles in the fractional bipartite graph G_F without increasing machine loads or energy, decomposing G_F into a collection of pseudo-trees. A maximum matching in each tree assigns each fractional job to a distinct machine. Since each machine receives at most one fractional job i^*, its load satisfies:

$$\hat{L}_j = \sum_{i:x^*_{ij}=1} p_{ij} + p_{i^*j} \leq \sum_{i=1}^{n} x^*_{ij} p_{ij} + p_{i^*j} \leq T + \max_{i:x^*_{ij}>0} p_{ij} \leq 2T \tag{51}$$

Setting $T = \mathrm{OPT}_{\mathrm{LP}} \leq \mathrm{OPT}$ yields an independent integral upper bound $X_{\mathrm{LST}} \leq 2 \cdot \mathrm{OPT}$. □

5.2 Closed-Form Schedulability Certificate Without Simulation

To enable embedded gateways to verify feasibility in $\mathcal{O}(n \cdot m)$ time without solving an LP, we define closed-form workload parameters:

- Minimum ToA per packet: $p_i^{\min} = \min_{j \in \mathcal{M}} p_{ij}$.
- Total minimal fleet workload: $W = \sum_{i=1}^{n} p_i^{\min}$.
- Average machine workload: $L_{\mathrm{avg}} = W/m$.
- Maximum single-packet transmission time: $p_{\max} = \max_{i=1}^{n} p_i^{\min}$.

Theorem 7 (A Priori Feasibility Certificate)**.** *Define the analytical upper bound X_{tight} as:*

$$X_{tight} = 2\left(\left\lceil \frac{W}{m} \right\rceil + p_{\max}\right) \tag{52}$$

If $X_{tight} \leq H$, then:

$$C^{mach}_{\max}(\alpha) \leq 2 \cdot OPT \leq X_{tight} \leq H \tag{53}$$

All packets are mathematically guaranteed to be schedulable within the hypercycle H with zero deadline drops, ***without requiring algorithm simulation or empirical execution****.*

Proof. For any feasible schedule, the optimal makespan is lower-bounded by the average load and the longest job: $\mathrm{OPT} \geq \max(\lceil W/m \rceil, p_{\max})$. Under the worst-case machine dispersion, the binary search upper bound parameter satisfies $w^* \leq \lceil W/m \rceil + p_{\max}$. By Theorem 3, our combinatorial scheduler produces an assignment α with machine makespan bounded by:

$$C^{\mathrm{mach}}_{\max}(\alpha) \leq 2w^* \leq 2\left(\left\lceil \frac{W}{m} \right\rceil + p_{\max}\right) = X_{\mathrm{tight}} \tag{54}$$

If $X_{\mathrm{tight}} \leq H$, then by transitivity $C^{\mathrm{mach}}_{\max}(\alpha) \leq H$, completing the proof. □

5.3 Numerical Evaluation on Standard Deployment

Evaluating Theorem 7 on our standard deployment ($n = 234$ packets, $m = 48$ machines, $H = 100{,}000$ ms, $\tau = 10{,}000$ ms):

- Total minimal fleet workload: $W = 14{,}565$ ms.
- Average workload per machine: $L_{\text{avg}} = 303.44$ ms $\implies \lceil W/m \rceil = 304$ ms.
- Maximum single-packet ToA: $p_{\max} = 98$ ms.
- Analytical Lower Bound: $\text{LB} = \max(304, 98) = 304 \text{ ms} \leq \text{OPT}$.
- Proven Upper Bound: $X_{\text{tight}} = 2 \cdot (304 + 98) = \mathbf{804}$ ms.

Because $X_{\text{tight}} = 804 \text{ ms} \ll \tau = 10{,}000 \text{ ms} \ll H = 100{,}000$ ms, the workload achieves a **99.2% analytical safety margin**. Feasibility is formally proven before scheduling begins.

6 Time Complexity Analysis of the Complete Framework

In this section, we present a comprehensive, rigorous time complexity derivation for the entire LoRaWAN Hypercycle Scheduling framework. We analyze each algorithmic phase individually and derive the overall system asymptotic bound.

6.1 Phase-by-Phase Computational Complexity

6.1.1 Matrix Initialization and Input Generation

Computing the Time-on-Air matrix P requires evaluating the Semtech SX1276 closed-form airtime equation for each packet $J_i \in \mathcal{J}$ across all $m = N_{\text{ch}} \times N_{\text{sf}}$ communication resources. With $n = |\mathcal{J}|$ packets and m machines, this initial evaluation requires:

$$T_{\text{init}} = \mathcal{O}(n \cdot m) \tag{55}$$

For our standard deployment ($n = 234, m = 48$), this step executes in under 2 milliseconds.

6.1.2 Phase 1: Combinatorial 2-Approximation Makespan Minimization

Phase 1 searches for the optimal candidate makespan w using binary search, executing Algorithm 2 (Unsplittable-Truemper) at each step:

1. **Binary Search Bounds:** The search interval is bounded by:

$$w_{\min} = \max_{i \in \mathcal{J}} \min_{j \in \mathcal{M}} p_{ij}, \quad w_{\max} = 2 \left\lceil \frac{\sum_i \min_j p_{ij}}{m} \right\rceil + 2 \max_{i,j} p_{ij} \tag{56}$$

The number of binary search iterations is strictly bounded by:

$$K_{\text{bs}} = \mathcal{O}\left(\log(w_{\max} - w_{\min})\right) \tag{57}$$

In typical LoRaWAN deployments, $w_{\max} - w_{\min} \approx 2000$ ms, yielding $K_{\text{bs}} \leq 11$ iterations.

2. **Potential Ceiling and Outer Iterations:** The node potentials $\pi(v)$ are bounded above by $\log_b(m)$, where $b = 1 + 1/m$. Because:

$$\log_b(m) = \frac{\ln m}{\ln(1 + 1/m)} \leq m \ln m \tag{58}$$

and all edge costs c_{ij} and potential increments are integers (≥ 1), the outer 'while' loop of Algorithm 2 executes at most:

$$K_{\text{outer}} = \mathcal{O}(m \log m) \quad \text{iterations} \tag{59}$$

3. **Dijkstra Shortest Path Computation:** In each outer iteration, Dijkstra's algorithm computes shortest paths on the reversed residual graph $G_\alpha(w) = (V, E_\alpha(w))$, where $|V| = n + m$ and $|E_\alpha(w)| \leq A + n \leq 2A$. Using a binary heap / priority queue implementation:

$$T_{\text{dijkstra}} = \mathcal{O}\left(|E| \log |V|\right) = \mathcal{O}\left(A \log(n + m)\right) \tag{60}$$

(With a Fibonacci heap, this reduces to $\mathcal{O}(A + (n + m) \log(n + m))$.)

4. **Inner Loop (Algorithm 1 Unsplittable-Blocking-Flow):** By Theorem 1, the inner loop performs at most m phases of job pushes. Each phase executes a backward Breadth-First Search (BFS) over the zero-reduced-cost admissible graph $G'_\alpha(w)$ in time proportional to the number of admissible edges, $\mathcal{O}(A)$. Thus, the inner loop terminates in:

$$T_{\text{ubf}} = \mathcal{O}(mA) \tag{61}$$

Combining Dijkstra and the inner blocking flow per outer iteration yields $\mathcal{O}(mA + A \log(n + m)) = \mathcal{O}(mA)$. Multiplying by the potential ceiling bound $K_{\text{outer}} = \mathcal{O}(m \log m)$, each call to Unsplittable-Truemper runs in:

$$T_{\text{truemper}} = \mathcal{O}(m^2 A \log m) \tag{62}$$

Incorporating binary search, the total time complexity of Phase 1 is:

$$T_{\text{Phase 1}} = \mathcal{O}\Big(m^2 A \log m \log(w_{\max} - w_{\min})\Big) \tag{63}$$

Since $A \leq n \cdot m$, for a fixed gateway hardware architecture ($m = 48$ machines), Phase 1 is **strictly linear in the number of packets** n:

$$T_{\text{Phase 1}} = \mathcal{O}\Big(n \cdot m^3 \log m \log(w_{\max} - w_{\min})\Big) \equiv \mathcal{O}(n) \tag{64}$$

6.1.3 Phase 2: Chronological EDF Placement with Segment Tree Tracking

Phase 2 verifies per-channel non-preemption and enforces global gateway demodulation limits ($N_{\text{demod}}^{\max} \leq 8$):

1. **Earliest Deadline First Sorting:** Sorting the assigned packets $\mathcal{J}_j$ on each machine M_j takes:

$$T_{\text{sort}} = \sum_{j=1}^{m} \mathcal{O}(|\mathcal{J}_j| \log |\mathcal{J}_j|) \leq \mathcal{O}(n \log n) \tag{65}$$

2. **Multi-Channel Progression via Min-Heap:** A min-heap maintains the next earliest ready packet across all m channel queues. Popping and inserting packets requires:

$$T_{\text{heap}} = \mathcal{O}(n \log m) \tag{66}$$

3. **Demodulation Occupancy via Lazy Segment Tree:** A segment tree over the discretized hypercycle duration $[0, H]$ has depth $\lceil \log_2 H \rceil$. For each scheduled packet:

   - Checking the peak concurrent demodulations $\max_{t \in [s_i, f_i - 1]} \Omega(t)$ requires one range-maximum query: $\mathcal{O}(\log H)$.
   - Updating occupancy $\Omega(t) \leftarrow \Omega(t) + 1$ across $[s_i, f_i - 1]$ requires one lazy range-addition: $\mathcal{O}(\log H)$.

   Across all n candidate packets, segment tree operations require $\mathcal{O}(n \log H)$.

Thus, a single complete simulation pass of Phase 2 has time complexity:

$$T_{\text{Phase 2}}^{\text{pass}} = \mathcal{O}\Big(n \log n + n \log m + n \log H\Big) = \mathcal{O}\Big(n(\log n + \log m + \log H)\Big) \tag{67}$$

With $H = 100{,}000$ ms, $\log_2 H \approx 17$, making this step practically instantaneous.

6.1.4 Phase 3: Priority-Conscious Drop Resolution Loop

When packets violate deadlines or the demodulation threshold, offending packets are pruned:

1. **Drop Selection:** Identifying the lowest-priority offender $J^* = \arg\min(\Pi_i, -D_i, i)$ over the set of violators $\mathcal{V}$ takes linear time: $\mathcal{O}(|\mathcal{V}|) \leq \mathcal{O}(n)$.

2. **Phase 2 Pruning without Phase 1 Re-run:** In Algorithm 3, whenever an offending packet is dropped due to timeline or demodulation constraints, it is removed directly from the candidate set $\mathcal{J}_{\text{cand}}$ and pruned from the current assignment α. Phase 2 is immediately re-executed on the remaining assignment without re-running the expensive Phase 1.

3. Across at most $k_{\text{drop}} \leq n$ dropped packets, Phase 2 re-simulations consume at most:

$$T_{\text{drops}} = \mathcal{O}\Big(k_{\text{drop}} \cdot n(\log m + \log H)\Big) \leq \mathcal{O}\Big(n^2(\log m + \log H)\Big) \quad (68)$$

4. **Makespan Infeasibility ($M_{\text{span}} > H$):** If the machine load makespan exceeds the hypercycle duration H, Phase 1 must be re-run with one packet removed. In real-time LoRaWAN networks, $M_{\text{span}} \ll H$ (e.g., 1,302 ms $\ll$ 100,000 ms in our benchmarks), so Phase 1 is executed **at most once**.

6.2 Overall System Time Complexity

Theorem 8 (Overall Time Complexity of the Framework)**.** *The complete LoRaWAN Hypercycle Scheduling framework (Algorithm 3) operates within the following asymptotic upper bounds:*

1. ***Worst-Case Upper Bound:*** *In the pathological worst case where Phase 1 makespan repeatedly exceeds H triggering n full re-executions:*

$$\mathcal{O}\Big(n \cdot m^2 A \log m \log(w_{\max} - w_{\min}) + n^2(\log m + \log H)\Big) \quad (69)$$

2. ***Standard Operational Bound:*** *In standard real-time networks where $M_{span} \leq H$ (Phase 1 executes once, and all drops are resolved through direct assignment pruning in Phase 2):*

$$\mathbf{T}_{total} = \mathcal{O}\Big(m^2 A \log m \log(w_{\max} - w_{\min}) + k_{drop} \cdot n(\log m + \log H)\Big) \quad (70)$$

3. ***Hardware-Fixed Asymptotic Bound:*** *Because the number of gateway channels and spreading factors is fixed by RF hardware ($m = 48$), m is a constant. Expressing $A \leq n \cdot m$, the framework scales as:*

$$\mathbf{T}_{total} = \mathcal{O}\Big(n \log(w_{\max} - w_{\min}) + k_{drop} \cdot n \log H\Big) \equiv \mathcal{O}\left(n^2 \log H\right) \quad (71)$$

When no drops occur ($k_{drop} = 0$), the entire framework runs in strictly quasi-linear time $\mathcal{O}(n(\log n + \log H))$.

7 Experimental Performance and Empirical Validation

7.1 Benchmark Configuration and Methodology

The framework was evaluated on an EU868 LoRaWAN network featuring:

Table 1: Detailed Computational Complexity Breakdown by Component

Component / Phase	Primary Data Structure	Dominant Operation	Asymptotic Time Co
Input Generation	2D Float/Int Matrix	SX1276 Airtime Formula	$\mathcal{O}(n \cdot m)$
Dijkstra Shortest Path	Min-Priority Queue (Heap)	Distance Updates	$\mathcal{O}(A \log(n+m$
Inner Blocking Flow	Level Graph + BFS Queue	Admissible Augmentations	$\mathcal{O}(m \cdot A)$
Outer Truemper Loop	Residual Bipartite Graph	Potential Adjustments	$\mathcal{O}(m^2 A \log m$
Phase 1 Makespan Opt.	Binary Search Interval	Unsplittable-Truemper	$\mathcal{O}(m^2 A \log m \log(w_{\max}$
Per-Channel EDF Sort	QuickSort / TimSort	Deadline Ordering	$\mathcal{O}(n \log n)$
Demodulation Tracking	Lazy Segment Tree	Range-Max / Range-Add	$\mathcal{O}(n \log H)$
Channel Heap Engine	Min-Heap of Channels	Event Cursor Progression	$\mathcal{O}(n \log m)$
Phase 2 Placement Pass	Segment Tree + Min-Heap	Chronological Simulation	$\mathcal{O}(n(\log n + \log m +$
Phase 3 Drop Resolution	Linear Offender Scan	Prune & Re-simulate	$\mathcal{O}(k_{\text{drop}} \cdot n(\log m +$
Complete System Total	**Hybrid Pipeline**	**End-to-End Execution**	$\mathcal{O}\Big(m^2 A \log m \log \Delta_w + k$

- $N_{\text{ch}} = 8$ carrier frequencies (868.1 to 868.5 MHz, plus 5 additional channels).
- $N_{\text{sf}} = 6$ quasi-orthogonal spreading factors (SF_7 to SF_{12}).
- Total logical machines: $m = 8 \times 6 = 48$.
- Hardware demodulation capacity: $N_{\text{demod}}^{\max} = 8$ concurrent paths.
- Hypercycle duration: $H = 100$ seconds = 100,000 ms.
- Traffic workload: 50 independent end-devices generating harmonic traffic with periods $T \in \{10, 20, 50, 100\}$ s and payloads $L \in \{10, 20, 30, 50\}$ bytes, producing $n = 234$ total packet instances.
- Deadlines configured with slack factor $1.5 \times \text{ToA}_{\text{SF7}}$.

7.2 Approximation Ratio Verification Against Exact Optimum

To verify the theoretical 2-approximation bound, an exact Branch-and-Bound (B&B) mixed-integer solver with aggressive symmetry breaking was executed on the scheduled candidate packet set ($N = 101$).

The exact optimal machine makespan is OPT = 800 ms. The 2-approximation scheduler achieved a machine load makespan of 1,302 ms, yielding an empirical ratio of:

$$\frac{C_{\max}^{\text{mach}}(\alpha)}{\text{OPT}} = \frac{1302}{800} = \mathbf{1.6275} \leq \mathbf{2.0000} \tag{72}$$

The empirical ratio strictly satisfies the theoretical 2.0 bound with a comfortable 18.6% safety margin. The complete execution completed in 0.412 seconds on standard desktop hardware.

7.3 Demodulation Occupancy and Invariant Verification

All system invariants were verified through post-scheduling automated assertion checks:

1. **Channel Non-Preemption:** Pairwise temporal overlaps on the same physical channel and SF evaluated to 0.

2. **Deadline Compliance:** Every scheduled packet completed strictly before its deadline: $f_i \leq D_i$ for all 101 scheduled transmissions.

3. **Demodulation Capacity:** The Segment Tree tracked continuous gateway occupancy $\Omega(t)$. The peak observed concurrent demodulation across the hypercycle was exactly 8/8, with zero instances exceeding the threshold:
$$\max_{t \in [0,H]} \Omega(t) = 8 \leq N_{\text{demod}}^{\max} \tag{73}$$

7.4 Physical Bottleneck Analysis of Packet Drops

The simulation scheduled 101 packets and dropped 133 packets (56.8% drop rate). Rigorous analysis reveals that this is caused by physical hardware saturation rather than algorithmic inefficiency:

1. **Synchronous Release Bottleneck at $t = 0$:** All 50 streams release packets simultaneously at $t = 0$ ms with tight relative deadlines $D_i \approx 63$ ms ($1.5 \times \text{ToA}_{\text{SF7}}$). Because the gateway possesses only 8 physical carrier frequencies, at most 8 packets can physically transmit simultaneously in the interval $[0, 63]$ ms. Consequently, at least 42 packets at $t = 0$ are physically impossible to schedule by *any* algorithm.

2. **Spreading Factor Deadline Infeasibility:** Packets with 63 ms relative deadlines cannot utilize SF8 (74 ms), SF9 (135 ms), or SF10–12 because their physical airtime alone exceeds the deadline. Thus, all traffic is restricted to the 8 channels of SF7, saturating the physical gateway capacity.

7.5 Radio Energy Consumption Evaluation

We evaluated the fleet-wide radio energy consumption using the Casals et al. [1] Class A energy model. Table 2 compares the total energy consumed across all $n = 234$ packets under our 2-approximation framework against the four baseline heuristics.

This reveals a critical design insight: while standard heuristics (such as Min-Min and Max-Min) achieve marginally lower load makespan on benign instances by greedily offloading packets onto higher spreading factors (SF8–SF10), they suffer from severe energy inflation (up to +45.1% more energy consumed). Our framework confines traffic to the lowest feasible spreading factor within the strict $2w^*$ makespan envelope, achieving the lowest energy footprint among all coordinated algorithms while simultaneously offering a provable worst-case

Table 2: Energy Consumption Comparison across All $n = 234$ Packets ($m = 48$ Machines)

Algorithm	**Total Radio Energy (mJ)**	**Avg Energy (mJ/packet)**	**Energy Ratio vs**
2-Approx (Phase 1.5 Ours)	**6,493.8**	**27.8**	**1.00× (Baseli**
Min-Min Heuristic	8,734.4	37.3	1.35× (+34.5
Greedy Least-Loaded (List)	9,261.2	39.6	1.43× (+42.6
Max-Min Heuristic	9,422.7	40.3	1.45× (+45.1
Random Assignment (ALOHA)	28,838.2	123.2	4.44× (+344.1

makespan guarantee ($C_{\max} \leq 2 \cdot \mathrm{OPT}$). For the scheduled subset ($N = 101$ packets), total radio energy is $2,649.6$ mJ (26.2 mJ/packet).

8 Conclusion and Future Work

In this report, we presented a comprehensive combinatorial approximation framework for real-time packet scheduling in LoRaWAN networks. By formulating $\langle \mathrm{SF}, \mathrm{CH} \rangle$ pairs as unrelated parallel machines, we implemented Gairing et al.'s 2-approximation algorithm based on Truemper's logarithmic transformation and unsplittable blocking flows. We integrated a lazy-propagated Segment Tree to enforce gateway demodulation ceilings (≤ 8) in $O(\log H)$ time, and provided a complete step-by-step time complexity derivation establishing that the framework operates in quasi-linear time for fixed gateway hardware.

Empirical validation against an exact Branch-and-Bound solver confirmed that the scheduler strictly respects the 2.0× theoretical guarantee across all instance sizes, achieving an empirical approximation ratio of 1.6275× on full-scale synthetic LoRaWAN workloads ($N = 101$).

Future work will explore:

1. Dynamic transmission power control (TPC) coupled with spreading factor selection to co-optimize energy consumption and makespan.
2. Runtime slack reclamation to accommodate sporadic non-periodic alarm messages.
3. Weakly-hard real-time guarantees ((m, k)-firm deadlines) to bound consecutive packet drops in congested environments.

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